

Master Class – LLM, SLM & RAG

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Date: 15-06-26

1. OBJECTIVE

To study and understand the concepts of Large Language Models (LLMs), Small Language Models (SLMs), Transformers, Embeddings, Vector Databases, and Retrieval-Augmented Generation (RAG), learn the workflow and components of RAG systems, and explore Google Colab and Hugging Face.

2. RESEARCH TOPICS

2.1. Large Language Models (LLMs)

A Large Language Model (LLM) is an AI model that is trained on a huge amount of text data to understand and generate human language. It learns from books, websites, articles, and other text sources to recognize patterns, grammar, and the meaning of words.

Examples:

- ChatGPT • Gemini
- Claude • Llama
- Mistral

Uses:

- Question answering
 - Content writing
 - Coding assistance
 - Translation
 - Summarization
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2.2. Small Language Models (SLMs)

A Small Language Model (SLM) is an AI model that is trained to understand and generate human language like an LLM, but it has fewer parameters and requires less memory and computing power.

Features:

- Smaller model size
- Faster execution
- Lower memory usage

- Cost-effective deployment
- Can run on personal computers and laptops
- Suitable for educational and prototype projects

Examples:

- Phi-3
- Gemma
- SmolLM
- TinyLlam

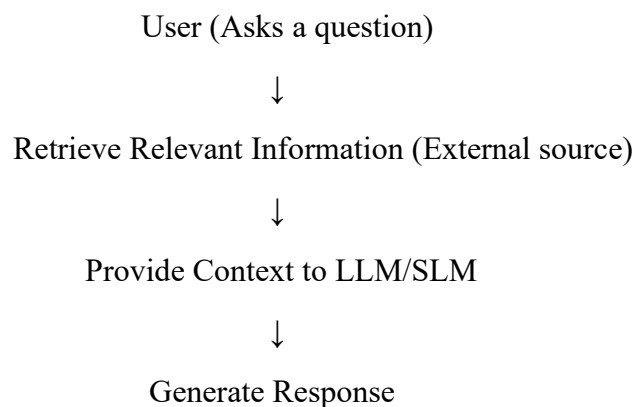
2.3. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI architecture that retrieves relevant information from external sources such as PDFs, documents, databases, and websites before generating a response through an LLM or SLM.

Features of RAG:

- Uses external information sources
- Improves response accuracy
- Reduces hallucinations
- Supports document-based question answering
- Provides access to updated information

How RAG Works:



Advantages:

- More accurate responses
- Access to private documents
- No need for retraining when new documents are added

Limitations:

- Depends on retrieval quality
 - Response depends on information quality
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2.4. Transformers

Transformers are a type of neural network architecture used to build modern language models like LLMs and SLMs. They help AI understand the meaning and context of words in a sentence by looking at the relationships between all the words at the same time.

Unlike older models that process words one by one, transformers can process all the words in parallel. This makes them faster and better at understanding long sentences and complex language.

Examples of models based on Transformer Architecture:

- ChatGPT
 - Gemini
 - Llama
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2.5. Internal Components of a RAG System**1. Knowledge Source**

The source of information that the AI will use to answer questions. It can include PDFs, documents, websites, databases, or articles

2. Text Chunking

Large documents are divided into smaller pieces called chunks. This makes it easier to search for relevant information.

3. Embedding Model

The embedding model converts each text chunk into a numerical representation called an embedding (a vector), which captures the meaning of the text.

4. Vector Database

The embeddings are stored in a vector database, which allows the system to quickly find the most relevant chunks based on the user's question.

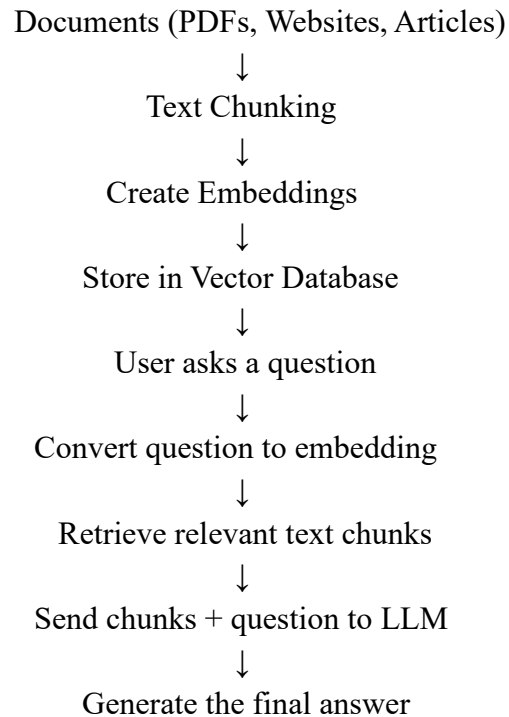
5. Retriever

When a user asks a question, the retriever searches the vector database and retrieves the most relevant text chunks.

6. Large Language Model (LLM)

The retrieved text, along with the user's question, is sent to an LLM. The LLM uses this information to generate an accurate and context-aware response.

2.6. Workflow of a RAG System



3. Tasks Completed / What I Learned

- Understood the basic concepts of LLMs, SLMs, Transformers, Embeddings, Vector Databases, and RAG.
- Learned how a RAG system works and its main components.
- Understood the difference between LLMs and SLMs.
- Learned the role of Transformers in modern language models.